EXPLORATORY DATA ANALYSIS OF GLOBAL TECH STARTUPS DATASET

Business Problem Statement

Between 2023 and 2026, tech startups faced tighter funding, falling valuations, widespread layoffs, and pressure to adopt AI. Investors, founders, and acquirers still lack a clear view of which startups are truly resilient and which are just running out of time.

This project analyzes startup data on funding, valuation, revenue, burn rate, runway, headcount, layoffs, AI adoption, and acquisition status. The goal is to find what separates startups that stayed strong from those that shrank, ran low on cash, or were acquired, so stakeholders can make better funding, hiring, and investment decisions.

SQL_Queries

```
--to check the complete dataset
select *
from global_tech_startups gts;
```

Company_ID	Domain	Founding_Year	Country	City	Funding_Stage	Total_Funding_USD_Millions	Valuation_USD_Billions	Annual_Revenue_USD_Millions	Monthly_Burn_Rate_USD_Millions	Runway_Months	Employee_Count	Layoffs_Last_12M	Revenue_Growth_Pct	Investor_Tier
B730AD52	HealthTech	2,014	Canada	Montreal	Seed	1.64	17.81	1.46	0.33	3.8	23	0	23	Tier 3
65A5F690	Web3/Crypto	2,014	China	Beijing	Pre-IPO	103.87	949.16	32.77	2.42	41.2	242	146	96	Tier 3
F106C853	Cybersecurity	2,020	USA	San Francisco	Series A	10.08	43.37	2.16	1.01	9.1	59	0	59	Tier 2
2BD798AF	FinTech	2,023	India	Delhi NCR	Series B	32.28	257.62	11.66	0.78	36.7	76	0	76	Tier 3
C5CB4451	Generative AI	2,020	USA	New York	Series C+	13.41	76.73	2.57	1.12	11	59	28	31	Tier 2
22994DE8	Generative AI	2,014	China	Shanghai	Series A	287.28	2,399.01	190.65	5	56.3	500	0	500	Tier 3
F998B9AE	Generative AI	2,017	USA	New York	Series A	6.55	72.71	6.22	0.61	9.2	32	0	32	Corp
DFEF57C1	EdTech	2,017	UK	Oxford	Seed	2.16	26.22	1.57	0.23	6.5	18	0	18	Tier 3
569C2230	FinTech	2,018	USA	San Francisco	Series C+	33.93	246.46	17.12	0.97	31.7	95	0	95	Tier 2

```
--check the total number of company
select 'number_of_rows', count(gts.Company_ID ) as 'values'
from global_tech_startups gts
union all
select 'number_of_company', count(distinct gts.Company_ID )
from global_tech_startups gts ;
```

'number_of_rows'	values
number_of_rows	25,000
number_of_company	24,999

```
--identify the duplicate row
select gts.Company_ID ,count(*) as NrOfCompany
from global_tech_startups gts
group by gts.Company_ID
order by count(*) desc;
```

Company_ID	NrOfCompany
7791E581	2
FFFA35D4	1
FFF0B0F3	1

Insight – there is a duplicate Company_ID

```
--confirm the duplicate row
SELECT *
from global_tech_startups gts
where gts.Company_ID = '7791E581';
```

Company_ID	Domain	Founding_Year	Country	City	Funding_Stage	Total_Funding_USD_Millions	12
7791E581	Generative AI	2,018	Israel	Jerusalem	Seed	7.56	
7791E581	HealthTech	2,020	USA	New York	Seed	0.72	

Insight - we discovered the company has the same Company_ID but different domain and different country which make us to assume that one of the Company_ID is a mistake

```
--check for empty cell
select *
```

```sql
from global_tech_startups gts
where Company_ID is null;
```

Company_ID	Domain	Founding_Year	Country	City	Funding_St	Total_Funding_USD_Million	Valuation_USD

Insight – there is no empty row in the dataset

```sql
--explore all the column to see if there are dimensions and measures
select gts.Company_ID
from global_tech_startups gts
limit 10;
```

	Company_ID
1	B730AD52
2	65A5F690
3	F106C853
4	2BD798AF
5	C5CB4451
6	22994DE8
7	F998B9AE
8	DFEF57C1
9	569C2230
10	859D4768

```sql
select distinct gts."Domain"
from global_tech_startups gts ;
```

	Domain
1	HealthTech
2	Web3/Crypto
3	Cybersecurity
4	FinTech
5	Generative AI
6	EdTech
7	ClimateTech
8	Robotics
9	SpaceTech
10	Autonomous Vehicles

Insight – there are altogether 10 domains to be considered in this dataset

```sql
select distinct gts.Founding_Year
from global_tech_startups gts ;
```

	Founding_Year
1	2,012
2	2,013
3	2,014
4	2,015
5	2,016
6	2,017
7	2,018
8	2,019
9	2,020
10	2,021

```sql
select min(gts.Founding_Year) as earliest_year
from global_tech_startups gts;
```

	123 earliest_year
1	2,012

```
select max(gts.Founding_Year) as latest_year
from global_tech_startups gts;
```

	123 latest_year
1	2,025

```
select distinct gts.Country
from global_tech_startups gts
order by gts.Country ;
```

	A-Z Country
1	Canada
2	China
3	France
4	Germany
5	India
6	Israel
7	Japan
8	Singapore
9	UK
10	USA

Insight – we have 10 countries to consider in the dataset.

```
select distinct gts.Country ,gts.City
from global_tech_startups gts
order by gts.Country ;
```

	A-Z Country	A-Z City
1	Canada	Montreal
2	Canada	Toronto
3	Canada	Vancouver
4	China	Beijing
5	China	Shanghai
6	China	Hangzhou
7	China	Shenzhen
8	France	Paris
9	France	Lyon
10	Germany	Berlin
11	Germany	Munich
12	India	Delhi NCR
13	India	Bengaluru
14	India	Mumbai
15	India	Hyderabad
16	Israel	Tel Aviv
17	Israel	Jerusalem
18	Japan	Kyoto
19	Japan	Tokyo
20	Singapore	Singapore
21	UK	Oxford

Insight – after classifying the locations by countries and cities, we have a total of 29 locations of tech startups.

```
select distinct gts.Funding_Stage
from global_tech_startups gts
order by
	case gts.Funding_Stage
	when 'Seed' then 1
	when 'Series A' then 2
	when 'Series B' then 3
```

```
        when 'Series C+' then 4
        when 'Pre-IPO' then 5
        when 'Post-IPO' then 6
        else 7
        end asc;
```

	Funding_Stage
1	Seed
2	Series A
3	Series B
4	Series C+
5	Pre-IPO
6	Post-IPO

```
select distinct gts.Investor_Tier
from global_tech_startups gts
order by
        case gts.Investor_Tier
              when 'Bootstrapped' then 1
              when 'Tier 1 (e.g. Sequoia, a16z)' then 2
              when 'Tier 2 (Mid-size VC)' then 3
              when 'Tier 3 (Angel/Seed)' then 4
              when 'Corporate VC (e.g. Google Ventures)' then 5
              else 6
        end asc;
```

	Investor_Tier
1	Bootstrapped
2	Tier 1 (e.g. Sequoia, a16z)
3	Tier 2 (Mid-size VC)
4	Tier 3 (Angel/Seed)
5	Corporate VC (e.g. Google Ventures)

```
select distinct gts.AI_Adoption_Level
from global_tech_startups gts
order by
        case gts.AI_Adoption_Level
              when 'Low' then 1
              when 'Medium' then 2
              when 'High' then 3
              else 4
        end asc;
```

	AI_Adoption_Level
1	Low
2	Medium
3	High
4	None

```
select distinct gts.Acquisition_Status
from global_tech_startups gts ;
```

	Acquisition_Status
1	Independent
2	Closed
3	Acquired
4	IPO
5	Acquired (Fire Sale)

```sql
--explore and calculate the KPI values
--total number of company
select count(*) as Number_of_company
from global_tech_startups gts
```

	123 Number_of_company
1	25,000

```sql
--Which countries and cities host the most startups
with startups as (SELECT
	gts.Country ,gts.City ,count(gts.Company_ID ) as NrOfStartups
from global_tech_startups gts
group by gts.Country ,gts.City )
select Country ,City ,NrOfStartups ,
	sum(NrOfStartups ) over (partition by country) as TotalStartups
from startups
order by TotalStartups desc;
```

	A-Z Country	A-Z City	123 NrOfStartups	123 TotalStartups
1	USA	Austin	1,776	11,178
2	USA	Boston	1,901	11,178
3	USA	Miami	1,844	11,178
4	USA	New York	1,893	11,178
5	USA	San Francisco	1,878	11,178
6	USA	Seattle	1,886	11,178
7	China	Beijing	921	3,755
8	China	Hangzhou	941	3,755
9	China	Shanghai	966	3,755
10	China	Shenzhen	927	3,755
11	India	Bengaluru	637	2,560
12	India	Delhi NCR	642	2,560
13	India	Hyderabad	631	2,560
14	India	Mumbai	650	2,560
15	UK	Cambridge	642	1,998
16	UK	London	691	1,998
17	UK	Oxford	665	1,998

Insight – top countries are USA, China, India, UK, Israel, Germany and Canada having more than 1000 startups.

```sql
-- How do funding, ARR and valuation change from Seed through Post-IPO
select
	gts.Funding_Stage,
	count(*) as NrOfCompany,
	round(avg(gts.Total_Funding_USD_Millions),1) as Avg_funding ,
	round(avg(gts.Revenue_ARR_Millions),1) as Avg_revenue,
	round(avg(gts.Valuation_USD_Millions),1) as Avg_valuation
from global_tech_startups gts
group by Funding_Stage
order by
	case Funding_Stage
		when 'Seed' then 1
		when 'Series A' then 2
		when 'Series B' then 3
		when 'Series C+' then 4
		when 'Pre-IPO' then 5
		when 'Post-IPO' then 6
		else 7
	end asc;
```

	Funding_Stage	NrOfCompany	Avg_funding	Avg_revenue	Avg_valuation
1	Seed	8,859	2.7	2.4	33.4
2	Series A	6,154	13.4	8.2	112.2
3	Series B	5,014	39.9	22.5	314.7
4	Series C+	2,996	108.9	61.2	835.6
5	Pre-IPO	1,257	287.8	157.3	2,127
6	Post-IPO	720	689.1	369.8	5,363.2

Insight – startups listed and about-to-be-listed in the stock market have more funding, revenue and valuation than early-stage startups.

```
--to see if tier investors add value to companies
select
	gts.Investor_Tier ,
	count(gts.Company_ID ),
	round(avg(gts.Total_Funding_USD_Millions),2) as Avg_funding ,
	round(avg(gts.Revenue_ARR_Millions),2) as Avg_revenue ,
	round(avg(gts.Valuation_USD_Millions),2) as Avg_valuation
from global_tech_startups gts
group by gts.Investor_Tier
order by
	case gts.Investor_Tier
		when 'Bootstrapped' then 1
		when 'Tier 1 (e.g. Sequoia, a16z)' then 2
		when 'Tier 2 (Mid-size VC)' then 3
		when 'Tier 3 (Angel/Seed)' then 4
		when 'Corporate VC (e.g. Google Ventures)' then 5
		else 6
	end asc;
```

	Investor_Tier	count(gts.Company_ID)	Avg_funding	Avg_revenue	Avg_valuation
1	Bootstrapped	1,175	0	1.83	25.52
2	Tier 1 (e.g. Sequoia, a16z)	2,544	118.54	69.17	916.27
3	Tier 2 (Mid-size VC)	7,477	53.77	30.13	417.05
4	Tier 3 (Angel/Seed)	9,965	56.11	30.44	438.48
5	Corporate VC (e.g. Google Ventures)	3,839	59.4	32.76	456.85

Insight – startups in tier 1 has most funding with the highest valuation.

```
--How many companies fall into each runway band (under 6, 6-12, 12-24 and 24+ months)
select
	CASE
		when gts.Runway_Months_2024 < 6 then 'Under 6'
		when gts.Runway_Months_2024 < 12 then 'Between 6 and 12'
		when gts.Runway_Months_2024 < 24 then 'Between 12 and 24'
		else '24+ months'
	END as Category,
	count(gts.Company_ID )
from global_tech_startups gts
group by Category ;
```

	Category	count(gts.Company_ID)
1	24+ months	2,868
2	Between 12 and 24	8,314
3	Between 6 and 12	5,592
4	Under 6	8,226

```
--tabular analysis by domain
select
	gts."domain" ,
	count(gts.Company_ID) as NrOfCompany ,
```

```
       round(avg(gts.Total_Funding_USD_Millions),2) as Avg_funding ,
       round(avg(gts.Revenue_ARR_Millions),2) as Avg_revenue ,
       round(avg(gts.Valuation_USD_Millions),2) as Avg_valuation,
       round(avg(gts.Runway_Months_2024),0) as Avg_runway
from global_tech_startups gts
group by gts."Domain"
order by Avg_valuation desc ;
```

	Domain	NrOfCompany	Avg_funding	Avg_revenue	Avg_valuation	Avg_runway
1	Generative AI	4,958	114.5	63.22	885.84	15
2	SpaceTech	1,252	97.8	50.31	747.87	14
3	Web3/Crypto	2,047	53.03	29	402.6	12
4	Autonomous Vehicles	1,248	46.65	26.37	376.42	11
5	FinTech	3,060	41.68	25.52	344.52	11
6	HealthTech	3,760	43.26	24.08	328.52	12
7	Robotics	2,468	41.92	23.34	323.08	11
8	ClimateTech	2,507	39.73	23.75	320.74	11
9	EdTech	1,234	40.89	20.53	298.67	11
10	Cybersecurity	2,466	36.55	21.38	292.3	11

Insight – this is the average analysis of all the startups based on domains which will help to filter the strong startups from the shrunk ones.

```
--tabular analysis by country
select
       gts.Country ,
       count(gts.Company_ID) as NrOfCompany ,
       round(avg(gts.Total_Funding_USD_Millions),2) as Avg_funding ,
       round(avg(gts.Revenue_ARR_Millions),2) as Avg_revenue ,
       round(avg(gts.Valuation_USD_Millions),2) as Avg_valuation
from global_tech_startups gts
group by gts.Country
order by Avg_valuation desc ;
```

	Country	NrOfCompany	Avg_funding	Avg_revenue	Avg_valuation
1	Japan	478	64.58	40.38	510.85
2	USA	11,178	61.84	35.05	491.03
3	Singapore	698	66.96	36.17	490.38
4	Germany	1,270	65.83	33.23	473.11
5	UK	1,998	56.04	31.63	443.37
6	Israel	1,277	59.56	30.85	442.54
7	China	3,755	56.04	31.68	435.86
8	France	729	57.97	32.96	432.69
9	India	2,560	56.2	30.08	432.37
10	Canada	1,057	50.84	29.64	394.04

Insight – all countries are around the same average so investing in any of the country is safe(location alone is a weak separator), besides the country with the highest valuation has little startups (Japan).

```
--Classify “stayed strong” vs “shrank”
SELECT
  CASE
    WHEN CAST(Current_Headcount_2026 AS REAL) / NULLIF(Peak_Headcount_2023, 0) >= 1   THEN
'Grew'
    WHEN CAST(Current_Headcount_2026 AS REAL) / NULLIF(Peak_Headcount_2023, 0) >= 0.8 THEN
'Stable'
    WHEN CAST(Current_Headcount_2026 AS REAL) / NULLIF(Peak_Headcount_2023, 0) >= 0.5 THEN
'Shrank'
    ELSE 'Shrank sharply'
  END AS Headcount_Trend,
  COUNT(*) AS NrOfCompany,
  ROUND(100.0 * COUNT(*) / SUM(COUNT(*)) OVER (), 1) AS Pct
FROM global_tech_startups
GROUP BY Headcount_Trend;
```

	Headcount_Trend	NrOfCompany	Pct
1	Grew	15,290	61.2
2	Shrank	5,842	23.4
3	Shrank sharply	2,892	11.6
4	Stable	976	3.9

Insight – many of the startups are growing.

```
--Do companies with deeper AI adoption retain more staff and lay off fewer
select
	gts.AI_Adoption_Level ,
	count(*),
	sum(gts.Layoffs_2024_2025 ) as Nr_of_LayoffStaffs,
	sum(gts.Peak_Headcount_2023 ) as Initial_nr_of_staffs,
	round(cast(sum(gts.Layoffs_2024_2025 ) as real)/cast(sum(gts.Peak_Headcount_2023 ) as
real)*100,2) ||'%' as layoff_rate
from global_tech_startups gts
group by AI_Adoption_Level
```

	AI_Adoption_Level	NrOfStartups	Nr_of_LayoffStaffs	Initial_nr_of_staffs	layoff_rate
1	High	9,814	307,884	2,034,756	15.13%
2	Low	5,082	158,630	1,040,582	15.24%
3	Medium	7,512	253,638	1,705,927	14.87%
4	None	2,592	68,931	512,233	13.46%

Insight – startups with higher AI-Adoption layoff more staffs but the layoff rate is still close to that of other startups

```
--potential companies we can invest
with potential_company_1 as (
select
	gts.Company_ID ,
	gts."Domain" ,
	gts.Country ,
	gts.Revenue_ARR_Millions ,
	gts.Valuation_USD_Millions ,
	gts.Funding_Stage ,
	gts.Acquisition_Status ,
	gts.AI_Adoption_Level ,
	gts.Runway_Months_2024
from global_tech_startups gts
where
	(gts.Acquisition_Status = 'Independent' or gts.Acquisition_Status = 'IPO')
	AND
	(gts.AI_Adoption_Level = 'High' or gts.AI_Adoption_Level = 'Medium')
	)
	select *,
		count(*) over () as NrOfCompany
	from potential_company_1 ;
```

Insight – with the formula above, we filtered out startups that are independent or listed in the stock market and also at least an intermediate level of AI-Adoption as the criteria 1 which short listed the startups to 11,133/25,000.

```
--companies that has higher potentials
with potential_company_2 as (
with potential_company_1 as (
select
	gts.Company_ID ,
	gts."Domain" ,
	gts.Country ,
	gts.Revenue_ARR_Millions ,
	gts.Valuation_USD_Millions ,
```

```
	gts.Funding_Stage ,
	gts.Acquisition_Status ,
	gts.AI_Adoption_Level ,
	gts.Runway_Months_2024
from global_tech_startups gts
where
	(gts.Acquisition_Status = 'Independent' or gts.Acquisition_Status = 'IPO')
	AND
	(gts.AI_Adoption_Level = 'High' or gts.AI_Adoption_Level = 'Medium')
	)
select *
from potential_company_1
where
	("Domain" = 'Generative AI' and potential_company_1.Valuation_USD_Millions >= 885 and
potential_company_1.Runway_Months_2024 >= 15)
	OR
	("Domain" = 'SpaceTech' and potential_company_1.Valuation_USD_Millions >= 747 and
potential_company_1.Runway_Months_2024 >= 14)
	OR
	("Domain" = 'Web3/Crypto' and potential_company_1.Valuation_USD_Millions >= 402 and
potential_company_1.Runway_Months_2024 >= 12)
	OR
	("Domain" = 'Autonomous Vehicles' and potential_company_1.Valuation_USD_Millions >=
376 and potential_company_1.Runway_Months_2024 >= 11)
	OR
	("Domain" = 'FinTech' and potential_company_1.Valuation_USD_Millions >= 344 and
potential_company_1.Runway_Months_2024 >= 11)
	OR
	("Domain" = 'HealthTech' and potential_company_1.Valuation_USD_Millions >= 328 and
potential_company_1.Runway_Months_2024 >= 12)
	OR
	("Domain" = 'Robotics' and potential_company_1.Valuation_USD_Millions >= 323 and
potential_company_1.Runway_Months_2024 >= 11)
	OR
	("Domain" = 'ClimateTech' and potential_company_1.Valuation_USD_Millions >= 320 and
potential_company_1.Runway_Months_2024 >= 11)
	OR
	("Domain" = 'EdTech' and potential_company_1.Valuation_USD_Millions >= 298 and
potential_company_1.Runway_Months_2024 >= 11)
	OR
	("Domain" = 'Cybersecurity' and potential_company_1.Valuation_USD_Millions >= 292 and
potential_company_1.Runway_Months_2024 >= 11)
	)
select *,
	count(*) over () as NrOfCompany
from potential_company_2
order by Revenue_ARR_Millions desc, Valuation_USD_Millions desc ;
```

Insight – startups that their valuation and runway months is above the average value in the domain analysis are separated from the criteria 1 as the criteria 2 which shortlisted the startups to 1,969/11,133.

```
--Which are the top 3 most resilient companies in each domain
with top_companies as (
with potential_company_2 as (
with potential_company_1 as (
select
	gts.Company_ID ,
	gts."Domain" ,
	gts.Country ,
	gts.Revenue_ARR_Millions ,
	gts.Valuation_USD_Millions ,
	gts.Funding_Stage ,
	gts.Acquisition_Status ,
	gts.AI_Adoption_Level ,
```

```
	gts.Runway_Months_2024
from global_tech_startups gts
where
	(gts.Acquisition_Status = 'Independent' or gts.Acquisition_Status = 'IPO')
	AND
	(gts.AI_Adoption_Level = 'High' or gts.AI_Adoption_Level = 'Medium')
	)
select *
from potential_company_1
where
	("Domain" = 'Generative AI' and potential_company_1.Valuation_USD_Millions >= 885 and
potential_company_1.Runway_Months_2024 >= 15)
	OR
	("Domain" = 'SpaceTech' and potential_company_1.Valuation_USD_Millions >= 747 and
potential_company_1.Runway_Months_2024 >= 14)
	OR
	("Domain" = 'Web3/Crypto' and potential_company_1.Valuation_USD_Millions >= 402 and
potential_company_1.Runway_Months_2024 >= 12)
	OR
	("Domain" = 'Autonomous Vehicles' and potential_company_1.Valuation_USD_Millions >=
376 and potential_company_1.Runway_Months_2024 >= 11)
	OR
	("Domain" = 'FinTech' and potential_company_1.Valuation_USD_Millions >= 344 and
potential_company_1.Runway_Months_2024 >= 11)
	OR
	("Domain" = 'HealthTech' and potential_company_1.Valuation_USD_Millions >= 328 and
potential_company_1.Runway_Months_2024 >= 12)
	OR
	("Domain" = 'Robotics' and potential_company_1.Valuation_USD_Millions >= 323 and
potential_company_1.Runway_Months_2024 >= 11)
	OR
	("Domain" = 'ClimateTech' and potential_company_1.Valuation_USD_Millions >= 320 and
potential_company_1.Runway_Months_2024 >= 11)
	OR
	("Domain" = 'EdTech' and potential_company_1.Valuation_USD_Millions >= 298 and
potential_company_1.Runway_Months_2024 >= 11)
	OR
	("Domain" = 'Cybersecurity' and potential_company_1.Valuation_USD_Millions >= 292 and
potential_company_1.Runway_Months_2024 >= 11)
	)
select
	Company_ID,
	"Domain",
	Funding_Stage,
	Valuation_USD_Millions,
	Runway_Months_2024,
	rank() over (partition by "Domain" order by Valuation_USD_Millions desc,
Runway_Months_2024 desc) as Company_rank
from potential_company_2)
select *
from top_companies
where company_rank <= 3;
```

Insight – this formula extracts top 3 rank from the dataset filtered by criteria 2 which shortlisted to 30 startups.